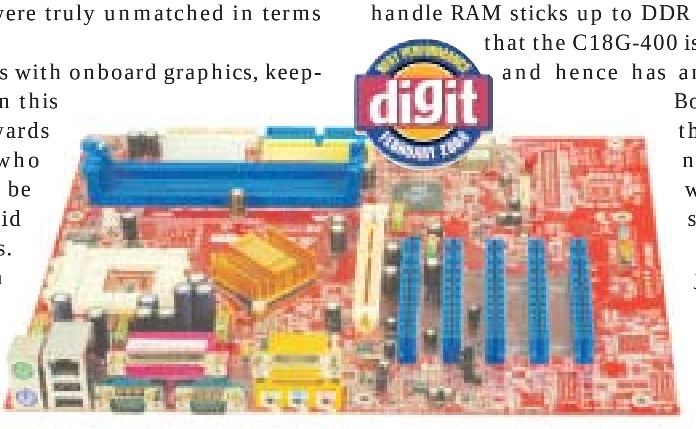
Most low-end motherboards for the AMD platform were based on VIA chipsets, ranging from the oldest KM266 to the latest KM400. Surprisingly, two boards based on the nForce2 chipset from Krypton made it to the low-end category due to their extremely low prices.

The boards based on the VIA KM266 chipset lacked many features. The boards did not support newer processors such as the Athlon XP 3200+, and memory standards such as the DDR 400 MHz—all of which reflected in their scores. The KM400 boards were at their best in terms of the features offered.

Performance-wise, the KT400 boards were no different from those based on the KM266, except for one that did manage to give those with the nForce2 chipset a stiff challenge. However, the nForce2 boards were truly unmatched in terms of performance and features.

We tested all low-end boards with onboard graphics, keeping in mind that most boards in this category were targeted towards price-conscious consumers who would expect the said feature to be present. However, two boards did not have integrated graphics. Hence, we tested them using a GeForce4 MX-440 display card. We plugged the same card into the board that topped the onboard graphics crowd. The scores thus obtained were then used to adjudge the winner.



The Krypton NF2400U isn't the most expensive board in this category, but it certainly outperforms the rest

Mercury's KT400FDSX was the lone board with the KT400—the predecessor of VIA's KM400 chipset. Its features are pretty much the same as those of the KM400 boards; the only exception being that it lacks an onboard graphics controller—the graphics needs are catered to by the 8X AGP slot. Also, this ATX motherboard has five PCI slots. However, despite the available extra space, the AGP-slot lock clearly interferes with the memory locks and is badly designed.

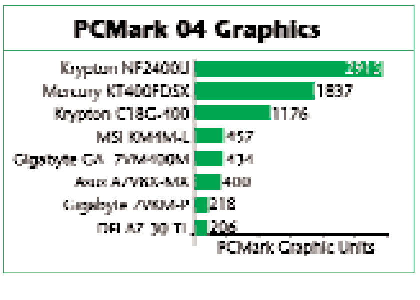
Krypton's C18G-400 and NF2400U, both based on the nForce2 chipset, were complete in terms of available features. Both support CPUs upto Athlon XP 3200+. While NF2400U supports a 400 MHz memory module, the C18G-400 can only handle RAM sticks up to DDR 333. One striking difference is that the C18G-400 is based on the IGP Northbridge, and hence has an integrated video controller. Both boards have an 8X AGP slot thus ensuring that you are not left stranded in case you want to upgrade your graphics sub-system.

While the C18G-400 has just 3 PCI slots, the NF2400U has five. Their layout design is relatively okay, but that of the NF2400U is better between the two. Overall, however, the Krypton C18G-400 walked away with the crown in the features section.

Features

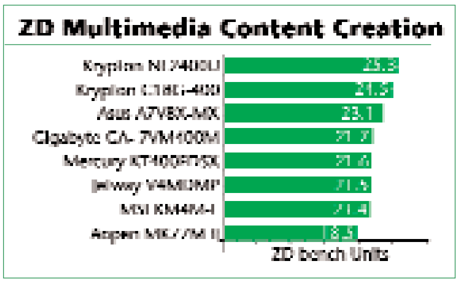
Boards such as the AOpen MK77M II, Gigabyte 7VKM-P and DFI AZ 30-TL based on the VIA KM266 chipset lack quite a number of features. The highest rated CPU they can support is the Athlon XP 2400+, running at 133 FSB. It can handle a maximum of DDR 266 MHz, thus limiting its upgradeability in the long run. They are courteous enough to provide a 4X AGP slot, but you might just not use it since many graphics cards have now migrated to 8X. The presence of six USB 2.0 ports, a LAN connector and onboard sound is quite a relief.

Coming to the KM400 boards such as the ASUS A7V8X-MX, Gigabyte's GA-7VM400M, Jetway's V4MDMP and MSI's KM4M-L, one sees a lot of upgrade options opening up. These boards can be plugged with an Athlon XP 2800+, and support DDR 333 MHz memory modules. However, the layout was cramped, courtesy the micro-ATX form factor. Nevertheless, MSI's KM4M-L and ASUS's A7V8X-MX have a reasonably good design. One aspect that makes them really worth their salt is the presence of the 8X-compliant AGP slots, apart from the integrated graphics. Thus, you have a choice to upgrade to a better graphics card when the need arises. Other features such as 6 USB 2.0 ports, integrated six-channel sound, onboard LAN and overclocking BIOS tweaks are just the norm.



Performance

As mentioned in the beginning, the boards with onboard graphics were compared to arrive at the winner. The winner was none other than the Krypton C18G-400. Not a single VIA chipset board could match up to this board's performance when tested for integrated graphics.



In the real-world tests, comprising the ZD Bench suite and the video-encoding test, the Krypton C18G-400 was way ahead of the competition. The only board that came close to its performance was the ASUS A7V8X-MX. The AOpen, DFI and Gigabyte's contenders could hardly stand up to the C18G-400's performance. The prime reason for such a difference was the memory sub-system.

While the nForce2 chipset-based C18G-400 relies on a better and faster memory standard, the other three still use older and slower memory standards, resulting in a performance drop even in applications used on a daily basis. This clearly indicates that the KM266 chipset is outdated—don't even consider using it for an office PC. On the other hand, the KM400 boards hovered around the 80 per cent mark but could not surpass the Krypton. Nevertheless, they still are in the race as shown by the ASUS A7V8X-MX board.